

Birla Institute of Technology & Science, Pilani
Work-Integrated Learning Programmes Division
First Semester 2018-2019
Comprehensive Examination (EC-3 Make-up)

Course No. : SS ZG526
 Course Title : DISTRIBUTED COMPUTING
 Nature of Exam : Open Book
 Weightage : 50%
 Duration : 3 Hours
 Date of Exam : 08/12/2018 (AN)

No. of Pages	= 6
No. of Questions	= 5

Note:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1 (a) Apply Singhal-Kshemkalyani's differential technique to determine the progress of vector clocks at each of the 4 processes shown in Fig. 1 and also the contents of the messages exchanged among them. Assume that initially the vector clocks of each of the 4 processes is $[0\ 0\ 0\ 0]$ and the increment value $d = 1$. [4]

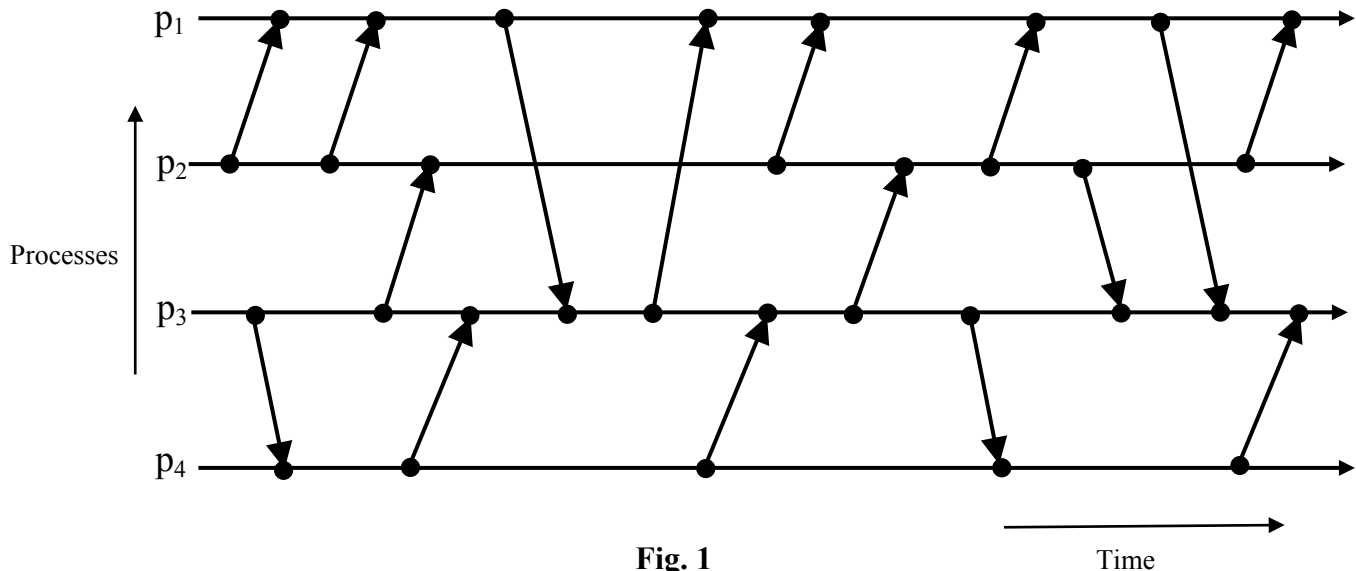


Fig. 1

Q.1 (b) Apply Fowler-Zwaenepoel's direct-dependency technique to determine the progress of vector clocks at each of the 4 processes shown in Fig. 1 and also the contents of the messages exchanged among them. Assume that initially the vector clocks of each of the 4 processes is $[0\ 0\ 0\ 0]$ and the increment value $d = 1$. [4]

Q.1 (c) Consider the Byzantine agreement scenario for a synchronous system shown in Fig. 2. Here, p_1 is the commander and p_2 , p_3 and p_4 are lieutenants. The messages that are exchanged between the processes are integers and can have two possible values – 4 and 8. The dashed arrows in the figure show the first round of message exchange and the labels along the arrows represent the values sent by the commander to the lieutenants. In this scenario, p_3 and p_4 are malicious processes and p_2 is a correct process. In the second round, p_3 sends false information to both p_2 and p_4 and p_4 sends false information to only p_2 . Show with the help of a diagram, the second round of message exchange among the lieutenants. Also, determine whether it is possible to reach a consensus in this scenario and substantiate your answer with proper reasoning. [1 + 1 = 2]

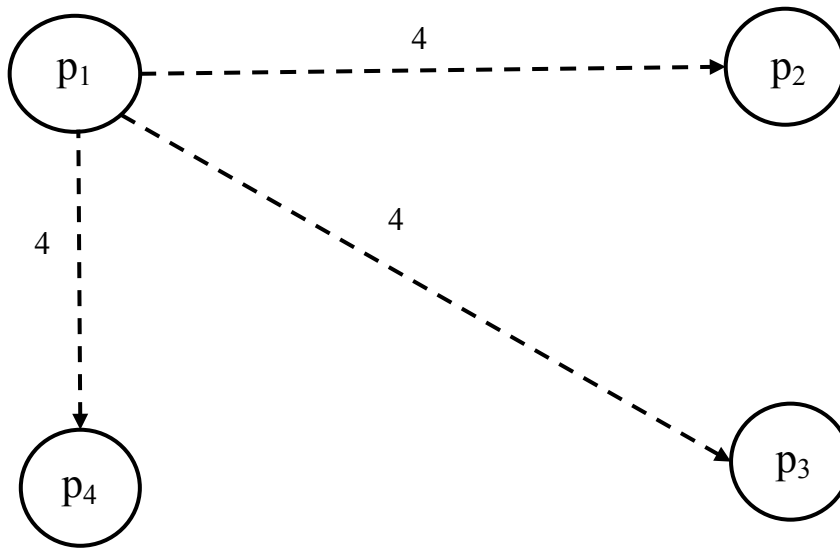


Fig. 2

Q.2 (a) Suppose a distributed system consists of 10 sites – $S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8, S_9$ and S_{10} . The request sets of S_1 and S_2 are as follows: $R_1 = \{S_1, S_3, S_4, S_6, S_8\}$ and $R_2 = \{S_2, S_4, S_8, S_9, S_{10}\}$. Both S_1 and S_2 wish to enter the critical section. Assume the following sequence of events:

- S_1 sends REQUEST(1) messages to S_3, S_4, S_6 and S_8 . S_2 sends REQUEST(2) messages to S_4, S_8, S_9 and S_{10} .
- S_4 receives the REQUEST(2) message before it receives the REQUEST(1) message.
- S_8 also receives the REQUEST(2) message before it receives the REQUEST(1) message.
- S_3 and S_6 receive the REQUEST(1) messages.
- S_9 and S_{10} receive the REQUEST(2) messages.

Using your knowledge of Maekawa's algorithm, describe the subsequent sequence of events until both S_1 and S_2 finish executing the critical section. Assume that no other critical section execution requests are pending, no other site wish to enter the critical section and a site does not send REQUEST, REPLY and RELEASE messages to itself. [6]

Q.2 (b) Consider a hypothetical distributed system consisting of 3 sites - S_1, S_2 and S_3 . This system uses a mutual exclusion algorithm where critical section execution requests are executed in order of the total execution time, i.e., the critical section execution request that takes a lesser amount of time to execute will get a chance to execute before a request which takes longer to execute. Here, execution time denotes the time interval between the entry and exit of critical section by a site. It is assumed that the critical section execution time of the sites are known beforehand. The timestamps of the requests play no role in determining the order of critical section execution. Now, suppose site S_2 has made a critical section execution request which will take 7 seconds to complete. In addition to this, sites S_1 and S_3 keep on continuously generating a series of critical section execution requests each of which will take 5 seconds to complete. Moreover, it is not known apriori that how many such critical section execution requests will be generated by S_1 and S_3 . Which property/properties of mutual exclusion algorithm is/are not satisfied in this scenario? Justify your answer. [1]

- Q.2 (c) Consider a synchronous distributed system consisting of 10 processes – $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9$ and P_{10} . Every process puts forth an initial integer value. The initial values put forth by $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9$ and P_{10} are 3, 1, 0, 8, 6, 7, 5, 3, 2 and 0 respectively. Out of the 10 processes, P_4, P_6 and P_9 are faulty processes and the non-faulty processes are aware of this fact. Using your knowledge of the interactive consistency problem, determine the values of the final agreement array. Assume that for a faulty process, the agreed upon value by all the non-faulty processes is 4. [0.5]
- Q.2 (d) Consider a peer-to-peer network which contains 30 objects. The total number of replicas for all the objects is 270. What is the number of replicas of each object under the uniform replication strategy? [0.5]
- Q.2 (e) Describe in detail how spanning-tree-based termination detection will be executed on the spanning tree of Fig. 3. Here, P_1 is the root node. [2]

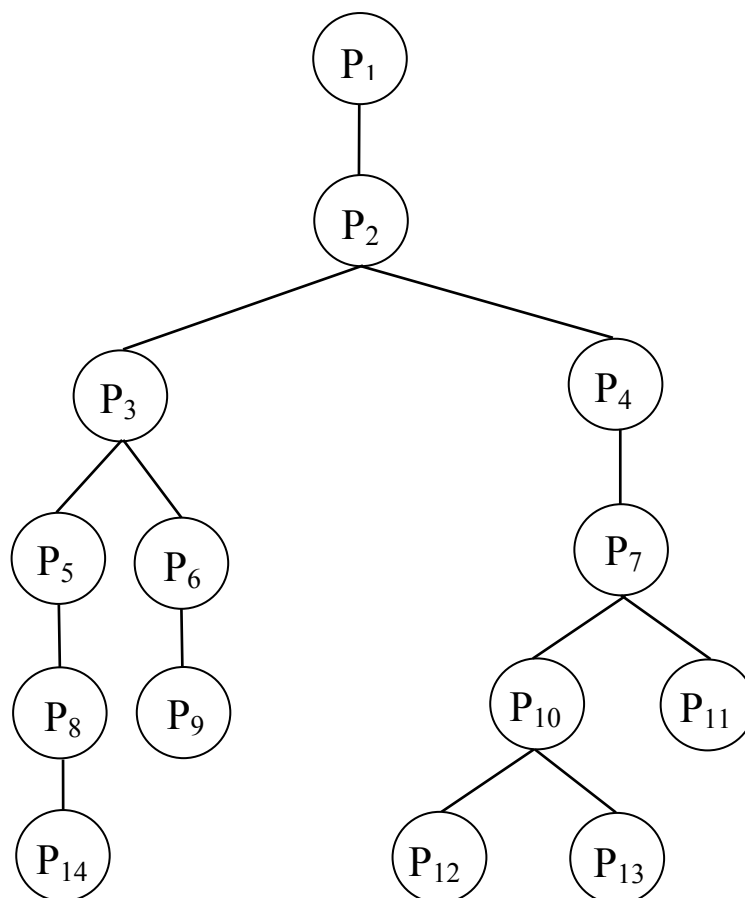


Fig. 3

- Q.3 (a) Consider a 3-stage Omega network consisting of 8 processors, 8 memory units and 12 switching elements each of size 2×2 . Between each pair of adjacent stages of the network, a link exists between output i of a stage and input j to the next stage according to the following interconnection function: $j = (5i + \log_2(n^2)) \% n$. Note that for this network, $n = 8$ and $0 \leq i \leq 7$. Draw a diagram depicting the interconnections between

The routing function from input line i to output line j of a particular stage is as follows: if the XOR of the leftmost and the rightmost bits of the binary representation of i is 0, the data is routed to the upper output wire, otherwise it is routed to the lower output wire. Using this information, determine for each input and output line combination in a particular stage, how data is routed in this Omega network. [4 + 4 = 8]

Q.3 (b) In a distributed system, there are 12 processes – $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}$ and P_{12} and 8 resources – $R_1, R_2, R_3, R_4, R_5, R_6, R_7$ and R_8 . Each resource has only one instance. If P_1 is waiting for both resources R_1 and R_2 to complete its execution, then the resource request of P_1 is expressed as R_1 *and* R_2 . If P_1 is waiting for either R_1 or R_2 , then the resource request is expressed as R_1 *or* R_2 . Using this notation, express the resource requests of the following processes as per the AND-OR model: [0.5 x 4 = 2]

- (i) P_2 is a blocked process and for completing its execution, P_2 requires either both of R_3 and R_4 or both of R_6 and R_7 .
- (ii) P_5 is a blocked process. For completing the execution, P_5 requires R_8 and any one of R_1, R_3 and R_4 and both of R_5 and R_6 .
- (iii) P_8 is a blocked process. In order to complete its execution, P_8 requires any one of R_1, R_3, R_4 and R_6 and both of R_7 and R_8 and any one of R_2 and R_5 .
- (iv) P_{10} is a blocked process. In order to complete the execution, P_{10} needs both R_2 and R_3 or any one of R_4, R_7 and R_8 and all of R_1, R_5 and R_6 .

Q.4 (a) Consider a distributed system consisting of 3 sites and 7 processes as depicted in Fig. 4.

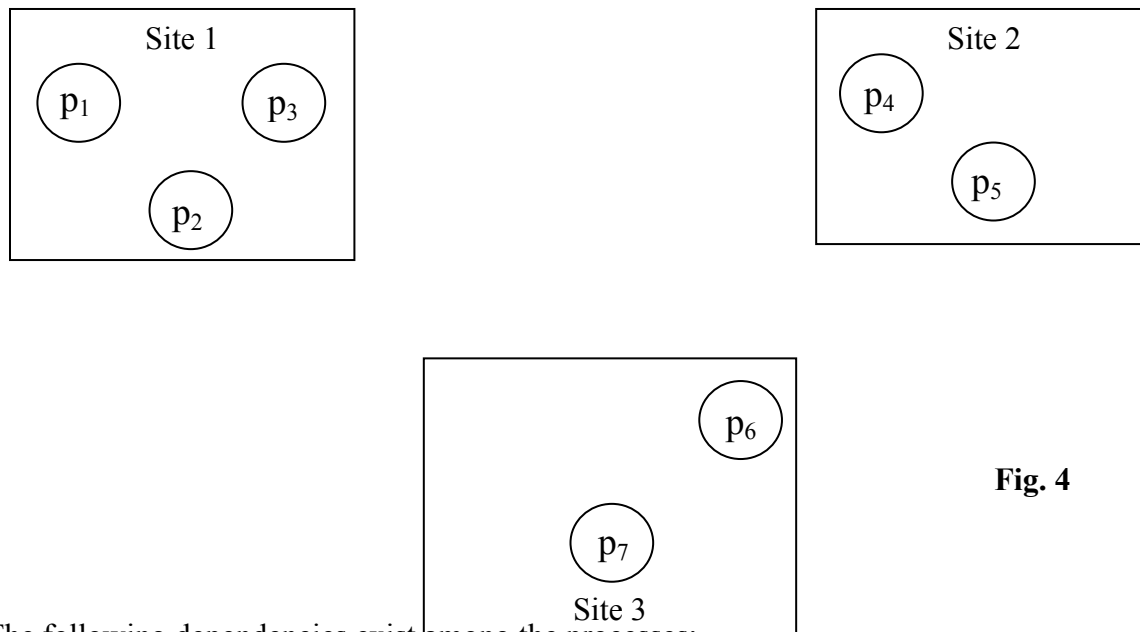


Fig. 4

The following dependencies exist among the processes:

- p1 is blocked and is waiting for p3 to release some resource
- p2 is blocked and is waiting for p3 to release some resource
- p3 is blocked and is waiting for p4 to release some resource
- p4 is blocked and is waiting for p6 to release some resource

- p_6 is blocked and is waiting for p_7 to release some resource
- p_7 is blocked and is waiting for p_3 to release some resource

Assuming all requests to be AND requests, execute Chandy-Misra-Haas algorithm for the AND model on this distributed system to determine whether any deadlock exists in the system. Assume p_3 initiates the deadlock detection and initially, the dependent arrays of all the processes contain false as values. Note that the deadlock is to be detected by using the algorithm and not by any other means. [4]

Q.4 (b) Suppose a distributed system consists of 5 sites – S_1, S_2, S_3, S_4 and S_5 . For every site, there is an incoming channel and an outgoing channel. The channel from site S_1 to S_2 is denoted as C_{12} and the channel from site S_2 to S_1 is denoted as C_{21} . The same applies for the remaining pairs of sites. What is the total number of marker messages required if the global snapshot of this system is to be recorded using the Chandy-Lamport snapshot recording algorithm? Give proper reason in support of your answer. [1]

Q.4 (c) Fig. 5 shows a distributed system consisting of 3 sites - S_1, S_2 and S_3 . Both S_2 and S_3 wish to enter the critical section and broadcast timestamped REQUEST messages corresponding to events a and b respectively. The timestamp of both events a and b is 2. Execute Ricart-Agrawala algorithm for implementing distributed mutual exclusion on this distributed system. Clearly show the exchange of REPLY messages among the sites, contents of request-deferred array of each of the sites corresponding to the send and receive events of the relevant messages and the events when each of the requesting sites enter and exit the critical section. [4]

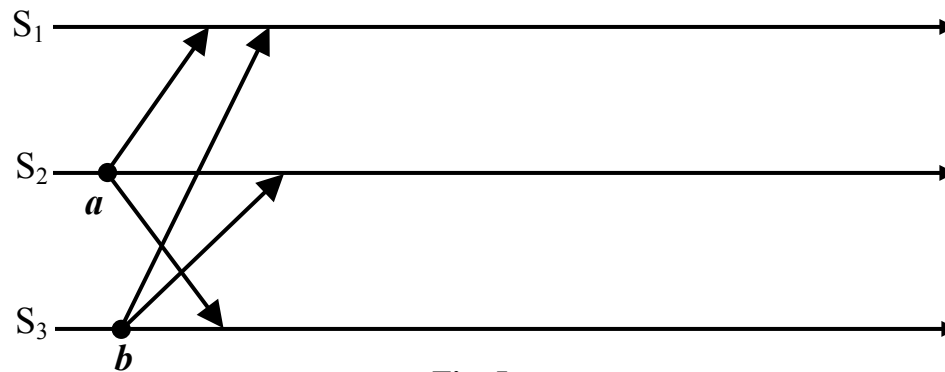


Fig. 5

Q.4 (d) For a 5-dimensional hypercube, determine the lengths of the shortest paths between the node pairs having the following labels. [0.5 + 0.5 = 1]

- (i) 01110 and 11100
- (ii) 10110 and 01111

Q.5 For a Chord ring of $m = 10$, consider the following arrangement of nodes and the placement of the key as shown in Fig. 6.

- (a) Determine the finger tables of N130, N410 and N654. [3 x 2 = 6]
- (b) The search for K880 is initiated at N130. Determine the step-by-step procedure by which the key is to be located using the scalable lookup algorithm. [4]

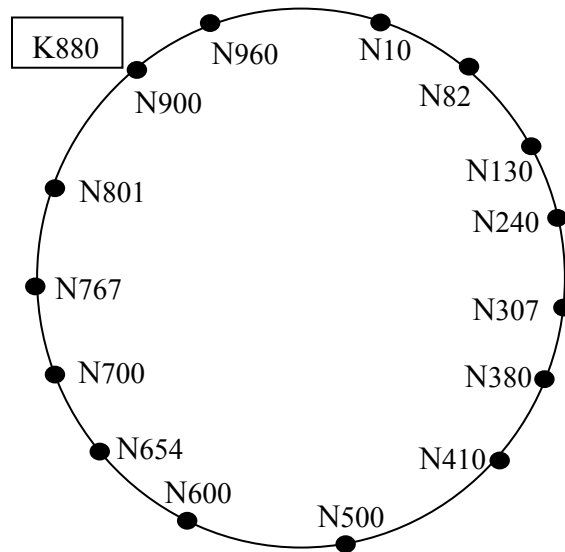


Fig. 6

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First Semester 2018-2019

Mid-Semester Test
(EC-2 Regular)

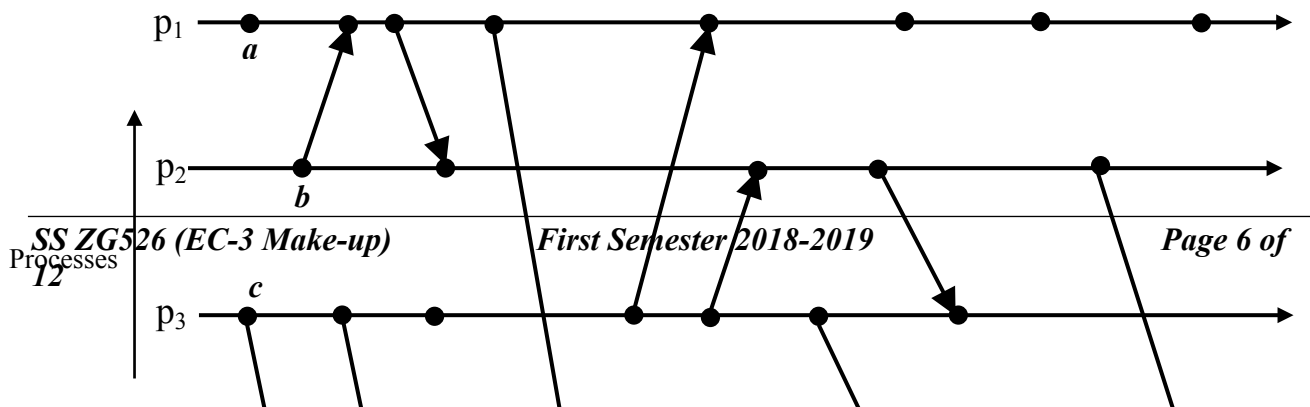
Course No. : SS ZG526
 Course Title : DISTRIBUTED COMPUTING
 Nature of Exam : Closed Book
 Weightage : 30%
 Duration : 2 Hours
 Date of Exam : 29/09/2018 (AN)

No. of Pages = 3
 No. of Questions = 5

Note:

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5. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
6. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1 (a) Consider the space-time diagram shown in Fig. 1. The scalar clock values of the events *a*, *b* and *c* are 2, 3 and 1 respectively and initially, the scalar clock of process *p*₄ is 0. Identify the scalar clock values of the events and the timestamps of the messages exchanged among the 4 processes. Use **R1** and **R2** for updating the scalar clocks. Assume *d* = 2. [5]



- Q.1 (b) Define open group and closed group multicast algorithms. [0.5 + 0.5 = 1]
- Q.2 (a) Explain termination detection using weight throwing [2]
- Q.2 (b) Describe the steps of Lamport's algorithm for distributed mutual exclusion. [3]
- Q.2 (c) Explain message queue using a diagram [1]
- Q.3 (a) Discuss the 3 properties of a synchronous system. [1]

- Q.3 (b) Describe the steps of the broadcast and the convergecast algorithms on the spanning tree shown in Fig. 2. [2 + 2 = 4]

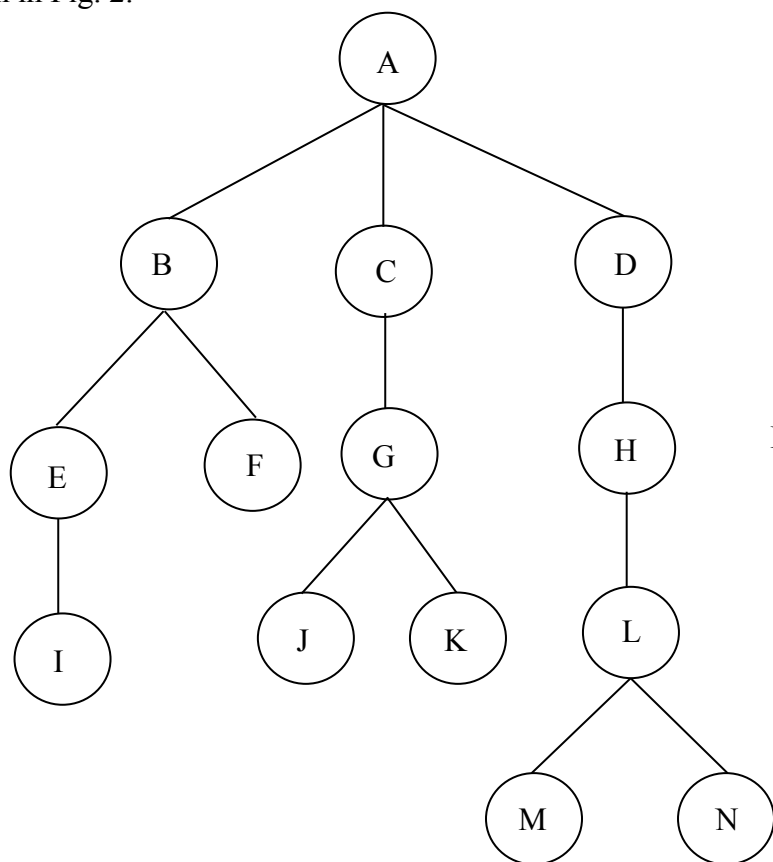


Fig. 2

- Q.4 (a) Consider the spanning tree shown in Fig. 3. Node N3 holds the privilege.

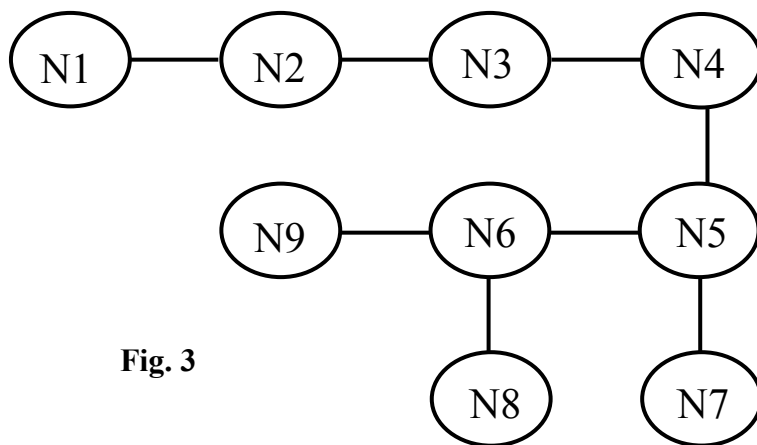


Fig. 3

Using the knowledge of Raymond's tree-based algorithm, answer the following questions.

- Determine the values of the HOLDER variables of each of the nodes. [3]
- Redraw the spanning tree of Fig. 3 by replacing the undirected edges with directed edges. [1]

- Q.4 (b) Explain isomorphism and event counting properties of vector time. [1 + 1 = 2]

- Q.5. Consider the money transfer scenario shown in the timing diagram of Fig. 4. S1, S2 and S3 are three sites of a distributed system which maintain bank accounts A, B and D respectively. C_{12} is the communication channel from site S1 to site S2, C_{21} is the communication channel from site S2 to site S1, C_{23} is the communication channel from site S2 to site S3, C_{32} is the communication channel from site S3 to site S2, C_{13} is the communication channel from site S1 to site S3 and C_{31} is the communication channel from site S3 to site S1. The dashed vertical lines denote the different time instants of this timing diagram. Initially, at time instant t_1 , account A has \$500, account B has \$800 and account D has \$300. Between time instants t_1 and t_2 , S1 initiates a transfer of \$20 from account A to account B. S2 receives this money transfer message after t_2 . Between time instants t_3 and t_4 , S1 initiates a transfer of \$30 from account A to account D. S3 receives S1's money transfer message after t_4 . Between time instants t_3 and t_4 , S3 initiates a transfer of \$80 from account D to account A. S1 receives S3's money transfer message after t_5 . These 3 money transfer messages are shown by bold arrows in the diagram. The amounts present in the 3 accounts at the different time instants are shown in the following table.

	t_1	t_2	t_3	t_4	t_5	t_6
A	\$500	\$480	\$480	\$450	\$450	\$530
B	\$800	\$800	\$820	\$820	\$820	\$820
D	\$300	\$300	\$300	\$220	\$250	\$250

Assume that the Chandy-Lamport algorithm is executed on this distributed system for recording the global snapshot. S1 initiates the snapshot recording algorithm by sending markers to S2 and S3 after time instant t_1 . S2 receives S1's marker between time instants t_1 and t_2 and sends a marker to S1 and S3. S1 receives S2's marker after t_3 and S3 receives S2's marker after t_3 . S3 receives S1's marker after t_2 and sends markers to S1 and S2. S2 receives S3's marker after t_3 and S1 receives S3's marker between t_4 and t_5 . The sending and the receiving of markers are shown using dotted arrows. Describe **how and when** the Chandy-Lamport algorithm records the local states of S1, S2 and S3 and the states of the channels C_{12} , C_{21} , C_{23} , C_{32} , C_{13} and C_{31} . Note that the recorded global state should be a consistent one. [7]

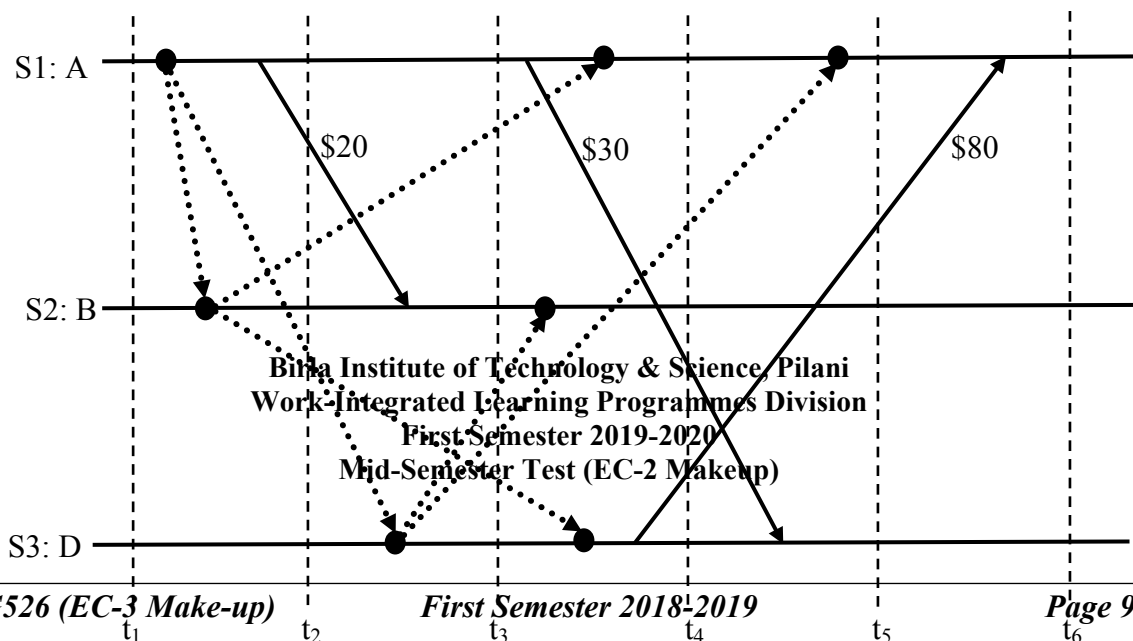


Fig. 4

Note:

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Q.1 (a) Grid computing is the collection of computer resources from multiple locations whereas cluster computing connects a number of readily available computing nodes via a fast local area network. How do they achieve scalability objective of distributed computing? Explain by giving examples.

[3]

Q.1 (b) Compatibility refers to the notion of interoperability among the resources in a distributed system. Which one (Binary level, Execution level or Protocol level) is more restrictive and why? Explain.

[2]

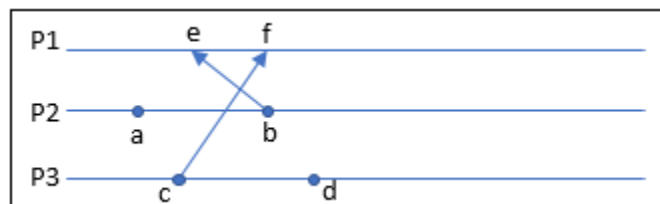
Q.1 (c) What are some of the important functions that a client stub and a server stub perform in the RPC model and why? Explain.

[2]

Q.2 (a) What is happened before relationship between events and how Lamport realized this? Describe with the help of an example.

[2]

Q.2 (b) Assign vector clock values to all the events shown in Fig.1 using the Fidge-Mattern's algorithm you studied in the course. Which events are concurrent in the given figure and why?



(Fig.1)

[2]

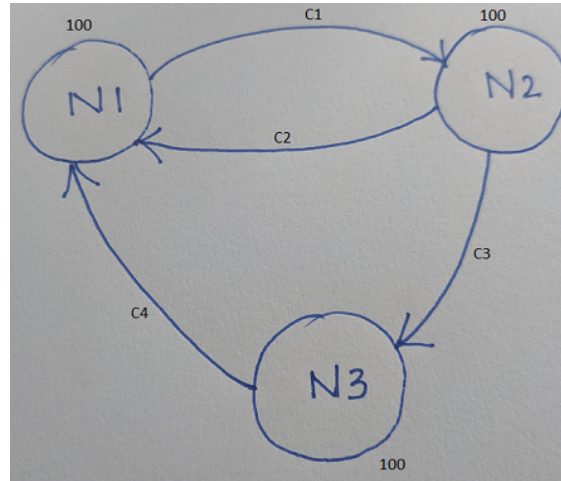
Q.2 (c) Assume that replicas of a data item are being kept at 3 different places in a distributed system. The replica managers at these places carry out updates that are received using messages from other sites. Give an example of a possible Causal Order (CO) violation in this scenario at one of the sites. What problem it (CO violation) might cause to the applications using these replicas? How do you overcome from this problem? Explain by giving a suitable example. You are free to use any protocol or algorithm that you have studied in the course.

[4]

Q.3 (a) Collected or recorded global states of a distributed computation may be used for purposes like detecting a distributed deadlock or detecting whether a distributed computation has terminated or not. Explain any other usage of this collected global state.

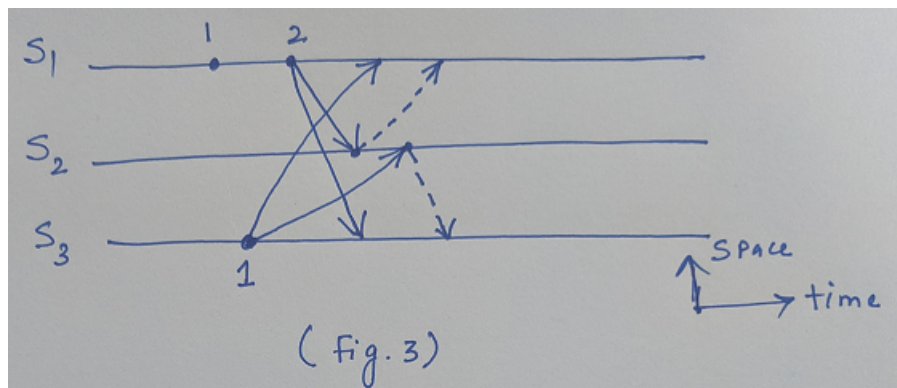
[2]

- Q.3 (b) Fig.2 shows a distributed application running at three nodes (N1, N2 and N3). Assume that communication channels (C1, C2, C3 and C4) are non-FIFO. Using Lai-Yang's algorithm, record the global state at time t4. Initial balances are as shown (100 at each site). The computations are listed in the chronological order as below:
 Time (t1): N1 sends 10 to N2 over C1, Time (t2): N2 sends 50 to N1 over C2, Time (t3): N3 sends 50 to N1 over C4, and Time (t4): N2 initiates the global state recording. Also, assume that the computation messages put over C1, C2 and C4 are yet to arrive at their respective sites.



(Fig.2)

- Q.4 (a) A Lamport's DME example is partly shown in Fig.3 with request messages (solid arrows), and few reply messages (dotted arrows). Complete the example with all remaining messages and find out the system throughput in the below case. Assume that the critical section execution time for both S1 and S3 is the same i.e. 10 seconds each and average message delay is 20 seconds. Show also the contents of the request_queue at each site.



(Fig. 3)

- Q.4 (b) Maewaka's request sets are generated using 4 conditions. What are those and what features (or benefits) these conditions bring in? Generate these request sets for N=7 using these four conditions. How does Maekawa's DME solve the problem of deadlocks? Give an example to demonstrate the solution.
